
```
clear all
close all
clc

%Equilibrium point
He = 20.6;
Le = 29.5;

%Desired Lynx Population
Ld = 40;

%Define initial conditions in z coordinates
H0 = 50;
L0 = 50;
z0 = [(H0-He), (L0-Le)];

%Open loop state space equations
A = [0.13 -0.93; 0.57 0];
B = [17.2 0]';
C = [0 1];
D = 0;

open_ss = ss(A,B,C,D);
open_ss_H = ss(A,B,[1 0],D); %Hare Output

t = 0:0.01:100; %Time Span
u = ones(size(t)); %Step Response

%Calculating Open Loop Response
[Z2o,T] = lsim(open_ss,u,t,z0);
[Z1o,T] = lsim(open_ss_H,u,t,z0);

%Shift Output from Z to X
Lo = Z2o + Le;
Ho = Z1o + He;

figure
plot(T,Lo)
hold on
plot(T,Ho)
xlabel('Time (Years)')
ylabel('Population')
legend('Lynxes', 'Hares')
axis([0 100 -100 200])
title('Open Loop Response')

%Lyapunov Stability Analysis
Q = eye(2);
P = lyap(A',Q);
lyapeig = eig(P);

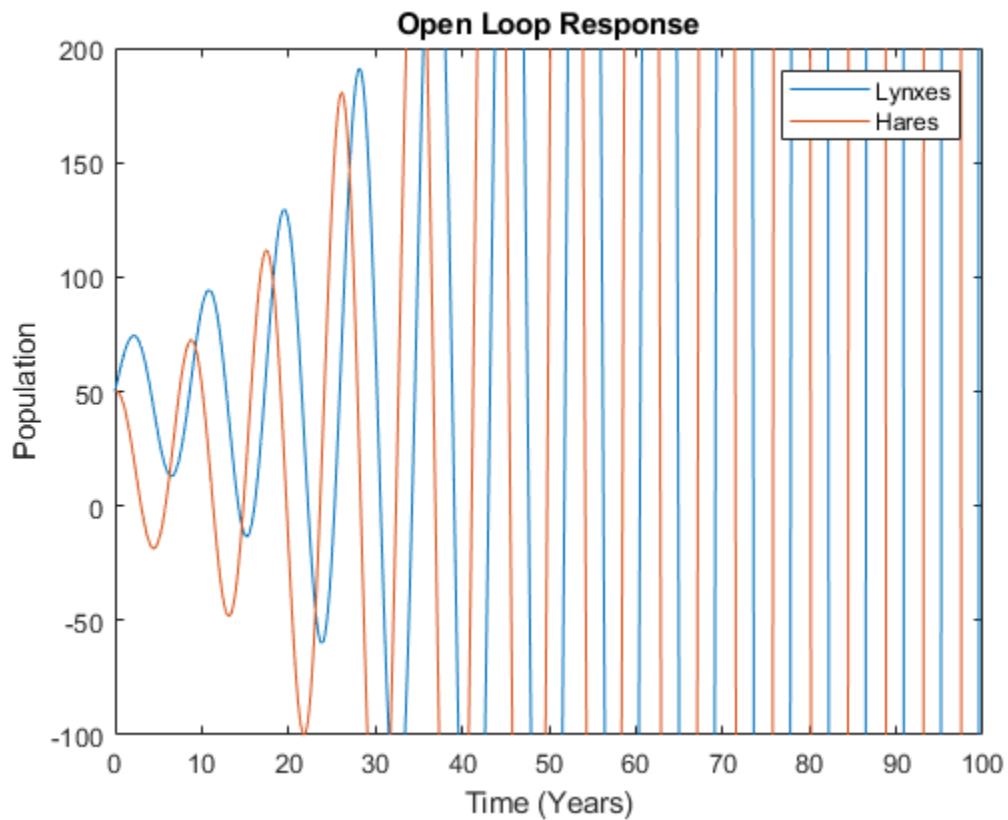
%Jordan Form
```

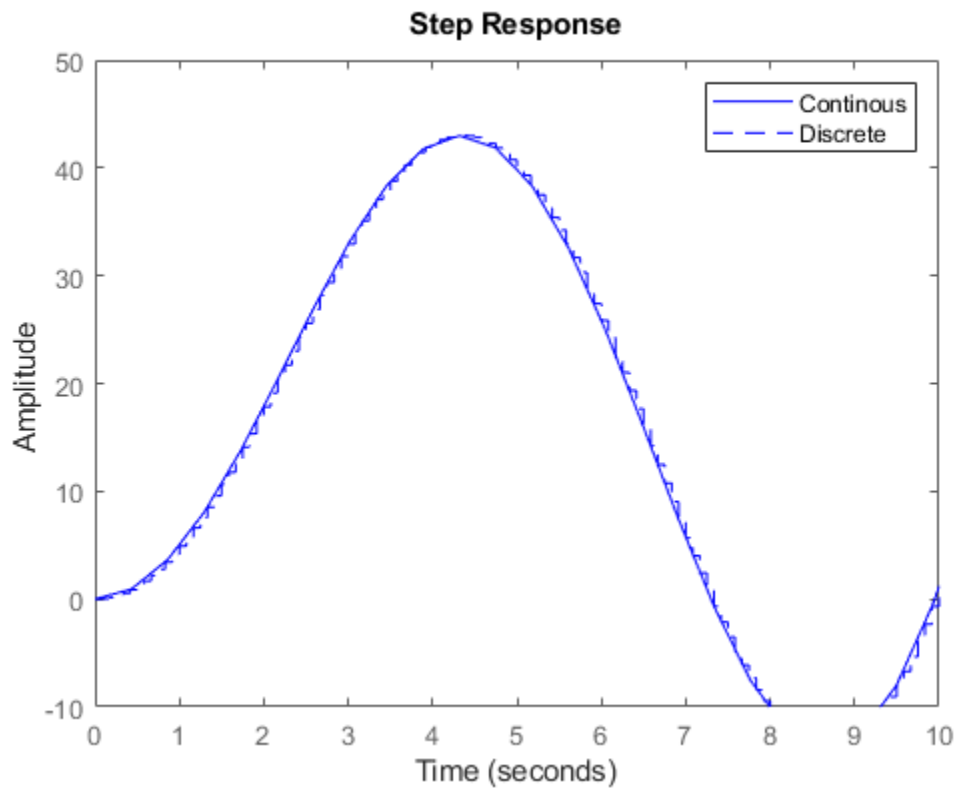
```
[V,J] = jordan(A);

%Open Loop Transfer Function
[z,p,k] = zpkdata(open_ss,'v');
Gs = zpk(z,p,k);

%Descretization and Subsequent Transfer Function
opend_ss = c2d(open_ss,1/12);
[zd,pd,kd,ts] = zpkdata(opend_ss,'v');
Gz = zpk(zd,pd,kd,ts);

figure
step(open_ss,'-',opend_ss,'--')
axis([0 10 -10 50])
legend('Continuous','Discrete')
```





State Feedback Control

```
%calculation of gain matrix and feedforward gain
```

```
K = place(A,B, [-0.1 -0.2]);
```

```
Q = C'*C;
```

```
Q(1,1) = 1.5;
```

```
R = 1;
```

```
%K = lqr(A,B,Q,R);
```

```
kf = -1/(C*inv(A-B*K)*B);
```

```
%Closed loop state space equation
```

```
Ac = A-B*K;
```

```
Bc = B*kf*(Ld-Le);
```

```
Cc = C;
```

```
Dc = D;
```

```
closed_ss = ss(Ac,Bc,Cc,D);
```

```
closed_ss_H = ss(Ac,Bc,[1 0],D);
```

```
%Calculating Closed Loop Response
```

```
[Z2,T] = lsim(closed_ss,u,t,z0);
```

```
[Z1,T] = lsim(closed_ss_H,u,t,z0);
```

```
%Shift output from Z to X
```

```
Lc = Z2 + Le;
```

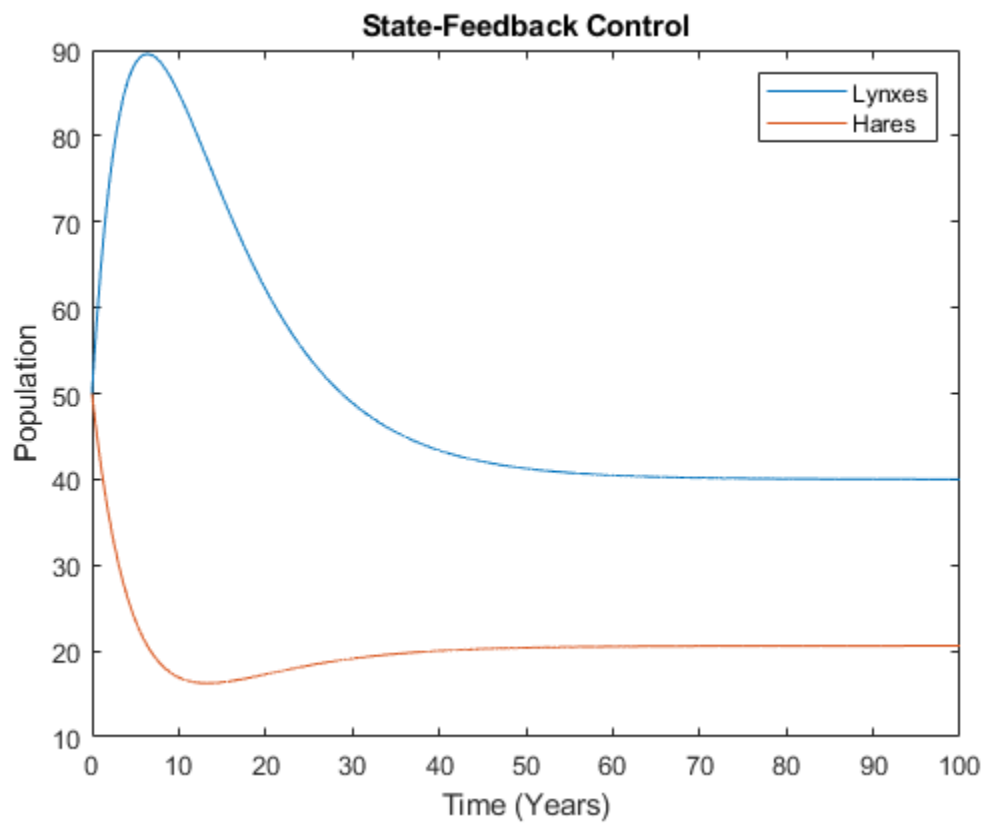
```
Hc = Z1 + He;
```

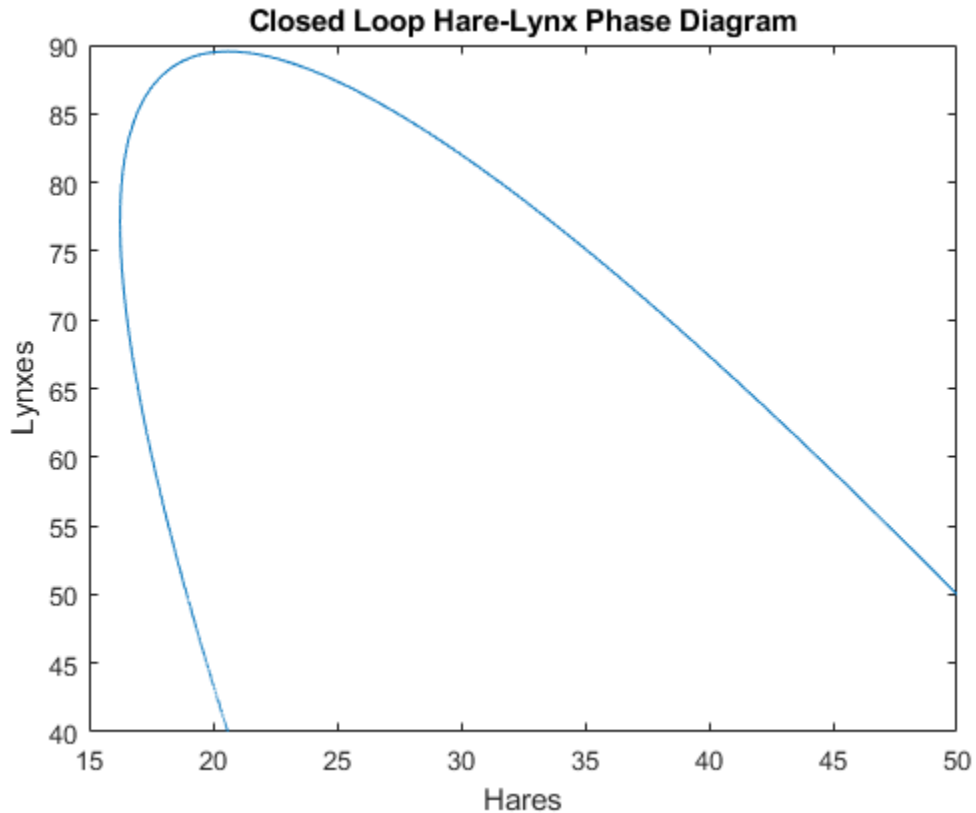
```

figure
plot(T,Lc)
hold on
plot (T,Hc);
xlabel('Time (Years)')
ylabel('Population')
legend('Lynxes','Hares')
title('State-Feedback Control')

figure
plot(Hc,Lc)
xlabel('Hares')
ylabel('Lynxes')
title('Closed Loop Hare-Lynx Phase Diagram')

```





Observability Based Control

```
P = [-1 -1.1];
L = place(A',C',P)';

%Observability state space equaton
Ace = [A-B*K, B*K; zeros(size(A)), A-L*C];
Bce = [B*kf*(Ld-Le); zeros(size(B))];
Cce = [0 1 0 0];
Dce = 0;
obs_ss = ss(Ace,Bce,Cce,Dce);
obs_ss_h = ss(Ace,Bce,[1 0 0 0],Dce);

z0obs = [z0 0 0];

%Calculating Observable Loop Response
[Z2obs,T] = lsim(obs_ss,u,t,z0obs);
[Z1obs,T] = lsim(obs_ss_h,u,t,z0obs);

%Shift Output from Z to X
Lobs = Z2obs + Le;
Hobs = Z1obs + He;

figure
plot(T,Lobs)
```

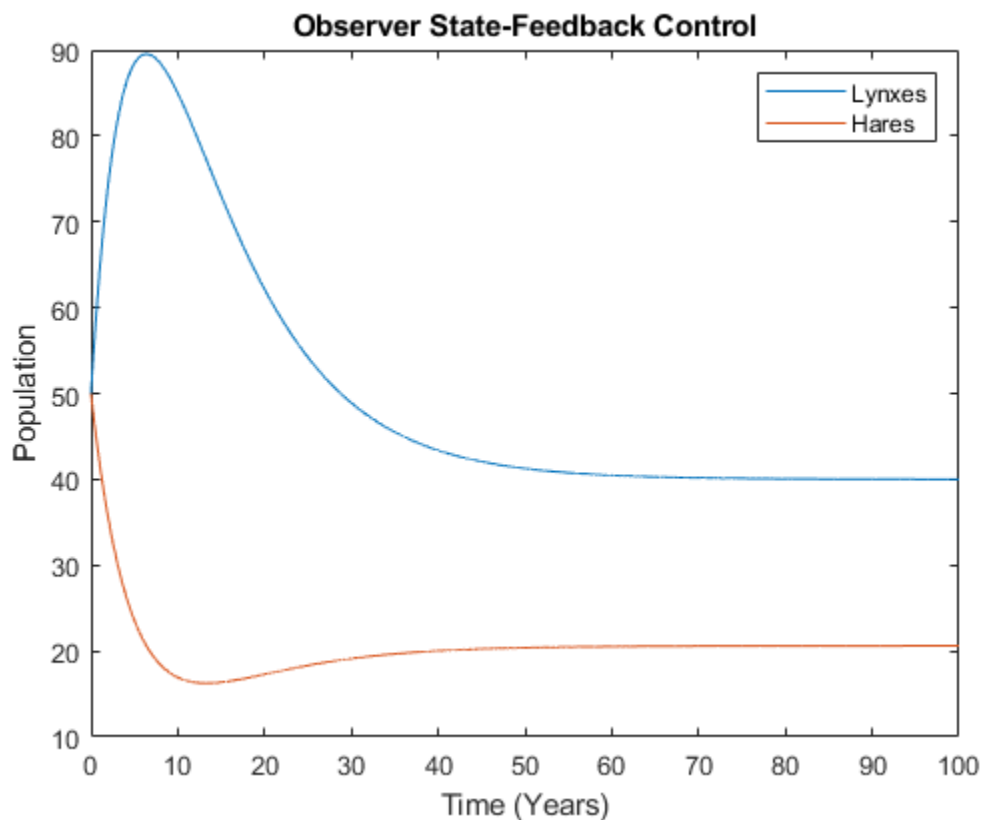
```

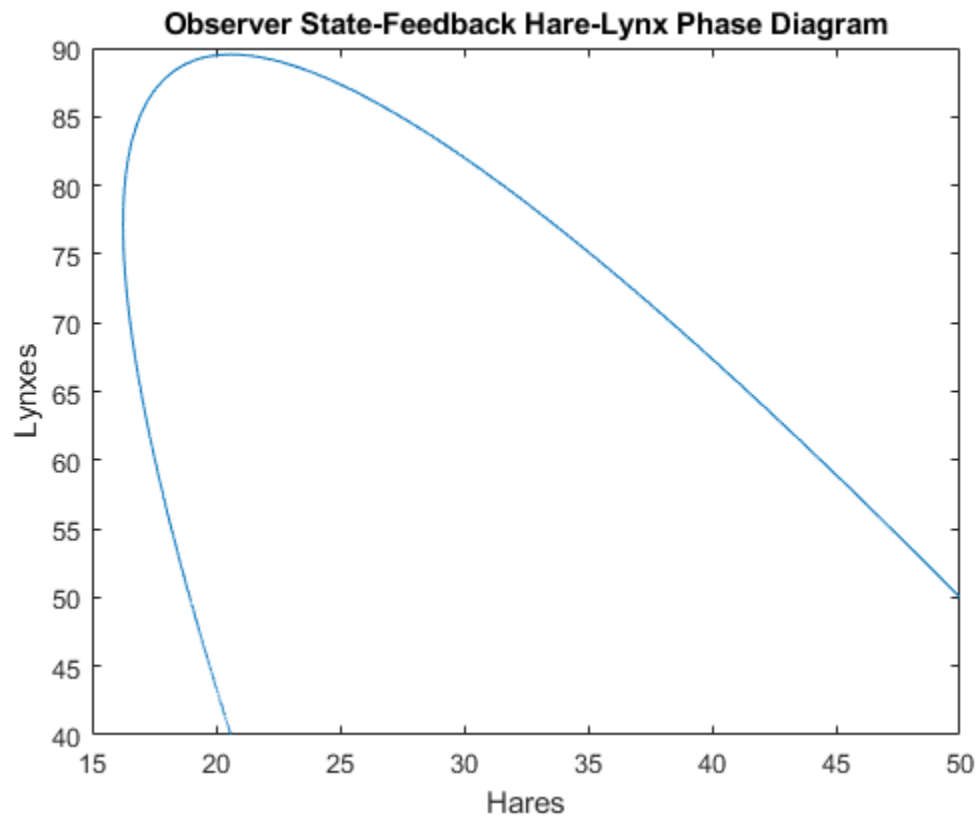
hold on
plot (T,Hobs);
xlabel('Time (Years)')
ylabel('Population')
legend('Lynxes','Hares')
title('Observer State-Feedback Control')

figure
plot(Hobs,Lobs)
xlabel('Hares')
ylabel('Lynxes')
title('Observer State-Feedback Hare-Lynx Phase Diagram')

%figure
%plot(T,Lc,'Linewidth',2)
%hold on
%plot(T,Lobs,'--','Linewidth',2)
%plot(T,Hc,'Linewidth',2)
%plot(T,Hobs,'--','Linewidth',2)
%xlabel('Time (Years)')
%ylabel('Population')
%legend('Lynx S-F','Lynx OS-F','Hare S-F','Hare OS-F')

```





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